

GEORGIA STATE
UNIVERSITY

LT 8440
e-Learning Environments

GROUP 5

Telu

Vimala

Colbrunn

Hailey

Streeter

A-Zahni

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WHO GETS IN?

A Critical Guide for Educators
Navigating VR/AR in Higher Education

Platform: Edcafe AI

Topic: VR/AR — Hope or Hype?

Audience: Higher Education Educators

Presentation: April 29, 2026



Presentation Agenda

What we will cover in 10 minutes

01 COMPONENT A

- ▶ The Problem We Solve
- ▶ Meet Vera — Our Chatbot
- ▶ Platform: Why Edcafe AI
- ▶ Theoretical Foundation
- ▶ Conversation Architecture
- ▶ CoI Framework — 4 Presences
- ▶ Engagement & Assessment
- ▶ Live Demo

02 COMPONENT B

- ▶ 5E Integration Framework
- ▶ WHAT — What Vera Does
- ▶ WHEN — Timing
- ▶ WHERE — Deployment
- ▶ WHY — Justification
- ▶ HOW — Educator Journey
- ▶ Design Rationale
- ▶ Thank You



The Problem — Why This Chatbot Exists

Higher education faculty need a critical framework, not more EdTech enthusiasm



THE HYPE

"VR/AR will transform education."

— Every EdTech conference, every year



THE REALITY

Faculty are pressured to adopt without a research-grounded framework to evaluate whether it genuinely fits their course and students.



OUR SOLUTION

Vera — a critical inquiry chatbot that guides educators through a personalized, evidence-based evaluation of VR/AR for their specific course.

Most faculty have no structured, research-grounded way to evaluate whether VR/AR makes sense for their specific course — Vera fills that exact gap.



Meet Vera — Your Skeptical-But-Curious Peer Mentor

“Not a VR headset. Not a tech evangelist. A skeptical-but-curious colleague who has read too many EdTech papers and wants to think through what they mean for YOUR classroom.”

The central question Vera helps educators answer:

“Should I use VR/AR in MY course — and will it work fairly for ALL my students?”

GROUNDING IN RESEARCH

9 research sources. Two PDFs uploaded to knowledge base: Hamilton et al. (2021) and Radianti et al. (2020).

COLLEGIAL IN TONE

Speaks as a peer, not an expert. Warm, honest, and occasionally surprising — never lecturing.

CRITICALLY HONEST

Tells educators when VR/AR is NOT the right tool for their course — that honesty builds trust.



Platform & Theoretical Foundation

Why Edcafe AI — and the research that shaped every design decision

WHY EDCAFE AI?

- ✓ **No login required**
Fully public link — meets rubric criterion 1
- ✓ **PDF knowledge base**
Vera draws on Hamilton (2021) + Radianti (2020) directly
- ✓ **QR code sharing**
Deployable instantly at faculty workshops
- ✓ **Voice capability**
Models our own UDL argument — multiple means of engagement
- ✓ **Chat tracking dashboard**
Facilitator can monitor all educator conversations

THEORETICAL FOUNDATION

Sfard (1998) — Participation Metaphor

Are students missing content or missing participation? VR/AR as a participation scaffold, not a delivery tool.

Vygotsky (1978) — Sociocultural Theory

VR/AR are cultural artifacts that reshape who participates and how — not neutral delivery mechanisms.

Lave & Wenger (1991) — Communities of Practice

Legitimate peripheral participation — VR/AR positions students inside disciplinary communities.

Garrison et al. (2000) — Community of Inquiry

Primary design framework — all four presences operationalized structurally throughout the chatbot.

Instructional Models Used

Scenario-based learning (equity scenario) + Problem-based learning (feasibility framework)



Conversation Architecture & Col Framework

Seven deliberate stages — four presences operationalized



COGNITIVE PRESENCE

Korea provocation = triggering. Myth/Reality = exploration. Equity scenario = integration. Postcard + Mirror Moment = resolution.
Practical Inquiry Model (Garrison et al., 2000)

SOCIAL PRESENCE

Named persona Vera — warm, collegial, honest. Confidence check creates psychological safety. Temperature check validates emotional responses.

TEACHING PRESENCE

One question per message. Deliberate 7-stage sequence. Explanatory feedback throughout. Branching adapts to each educator's concern — not a fixed script.

LEARNER PRESENCE

Educators choose discipline, name their concern, evaluate their own course, envision their future, and choose their own next step. Agency is structural.



Engagement Strategies & Assessment

Going beyond Q&A — Sections 3.5 and 3.6

4 ENGAGEMENT STRATEGIES

Reflective Prompts

Mirror Moment (opening), Postcard (closing), Mirror Moment revisited — a metacognitive arc across the full conversation.

Scenario-Based Learning

Prof. Rivera (nursing) + Prof. Osei (literature) — authentic professional decisions, no single correct answer.

Formative Check-ins

Temperature check mid-conversation (Hailey's suggestion) + Stage 5 biggest concern at key transition points.

Curated Resources

3 discipline-specific resources at closing — personalized to their subject and branching path.

5 EMBEDDED ASSESSMENTS WITH EXPLANATORY FEEDBACK

1

Myth or Reality — Round 1

Prediction + Hamilton et al. (2021) + Radianti et al. (2020)

2

Myth or Reality — Round 2

Accessibility claim + Rose & Meyer (2002) + CAST (2018)

3

Equity Scenario

4-option professional decision + non-judgmental research debrief

4

Feasibility Questions

3 open questions + honest verdict: strong, partial, or weak fit

5

Strategy Choice

First immersive step + one concrete next action this week

Live Demo — Walk Through With Us

app.edcafe.ai/chatbots/69efdb894eb81875b9dac2e2

SCAN OR VISIT



[app.edcafe.ai/chatbots/
69efdb894eb81875b9dac2e2](https://app.edcafe.ai/chatbots/69efdb894eb81875b9dac2e2)

WHAT WE WILL WALK THROUGH:

- 1 Korea provocation opens — triggering event (Garrison, 2000)
- 2 Mirror Moment — educator stakes a position before any info
- 3 Confidence check — learner presence from the start
- 4 Context capture — real course + challenge entered
- 5 Myth or Reality Round 1 — prediction before research feedback
- 6 Equity scenario — Option D chosen (Hailey's suggestion honored)
- 7 Temperature check — mid-conversation formative check-in
- 8 Branching Path C — equity and inclusion pathway activated

Feature Highlights — Instructional Purpose

Three Edcafe AI features — three design reasons (Component A requirement)



PDF Knowledge Base Upload

Technical: Two research PDFs uploaded into Edcafe AI — Hamilton et al. (2021) and Radianti et al. (2020). Vera draws on these throughout the conversation.

Instructional Purpose:

Grounds every response in peer-reviewed evidence rather than just our system prompt. Directly operationalizes Teaching Presence — research-backed scaffolding is visible in every stage.

Teaching Presence
(Garrison et al., 2000)



Voice Capability

Technical: Educators can listen to Vera rather than read. Audio available on every message through Edcafe's built-in voice feature.

Instructional Purpose:

We cannot advocate for Universal Design for Learning in our equity scenario while only offering a text experience. Voice models the very UDL principle we teach. Also supports Social Presence through vocal tone.

Social Presence +
UDL (CAST, 2018)



Teacher Chat Tracking Dashboard

Technical: Edcafe records every educator's full conversation and makes it visible to the facilitator in a tracking dashboard.

Instructional Purpose:

Allows an instructor or PD facilitator to monitor engagement patterns, identify common concerns, and adjust professional development design accordingly. Teaching Presence extends beyond the chatbot itself.

Teaching Presence
(Garrison et al., 2000)



5E Integration Framework — WHAT & WHEN

Component B: Situating Vera in the broader e-learning environment

W H A T

What is Vera designed to do?

Vera is a critical inquiry chatbot that tutors, scaffolds, assesses, and guides reflection — all personalized to each educator's real course.

It does what a static resource cannot: respond to what the educator actually says. A video delivers the same content to every viewer. Vera adapts every stage from Stage 2 onward to the educator's specific discipline, challenge, and learning goal.

Learning functions: tutoring, scaffolding, assessing, prompting reflection, and simulating collegial dialogue — all in one experience.

W H E N

When in the learning sequence is she used?

PRIMARY:

Pre-adoption professional development — BEFORE a faculty member commits to VR/AR implementation

SECONDARY:

Post-reading reflector — AFTER assigned literature on immersive technology

TERTIARY:

Asynchronous preparation — BEFORE a faculty workshop or department discussion

WHY THIS TIMING:

Maps to triggering + exploration phases of Garrison's Practical Inquiry Model (2000) — the critical decision window



5E Integration Framework — WHERE & WHY

Deployment context and justification grounded in theory

W H E R E

-  Embedded in iCollege or Canvas as a preparatory activity link
-  Shared via QR code at faculty development workshops — Edcafe generates automatically
-  Linked in department professional development newsletters or emails
-  Accessed independently — no login required on any device including mobile
-  Edcafe dashboard: facilitator monitors all conversations and identifies common concerns

W H Y

Why a chatbot and not a video, reading, or quiz?

Sfard (1998)

Learning that changes practice requires participation, not just exposure to information.

Vygotsky (1978)

A cultural tool that responds to you mediates reflection differently than a static text that does not.

Garrison et al. (2000)

Meaningful inquiry requires a triggering event, sustained exploration, and personal resolution. Vera scaffolds all three. A video cannot.

Pedagogical problem solved: one-size-fits-all professional development fails faculty. Vera personalizes every response to every educator's real course.

5E Integration Framework — HOW

The educator's actual journey from entry to exit

HOW does Vera support the learning experience in practice?



ENTERS

Provoked by Korea story — not welcomed. Puzzlement before any content.



REVEALS

Mirror Moment — stakes a position before receiving any information.



SHARES

Real course + challenge. Everything from here is personalized.



PREDICTS

Myth/Reality — predicts before getting research feedback.



DECIDES

Equity scenario — professional judgment under real constraints.



CHECKS IN

Temperature check — emotional + cognitive state validated.



NAMES

Biggest concern → routes to personalized branching path.



EVALUATES

3 feasibility questions applied to their actual course.



IMAGINES

Postcard — self-directed future visioning in 2-3 sentences.



CHOOSES

Strategy choice → 1 concrete next step this week.



REFLECTS

Mirror Moment revisited — own cognitive change made visible.



LEAVES

3 curated resources + Vera's final words. Exits with agency.



Design Rationale — Every Decision Traces to Theory

Rubric alignment across all five evaluation criteria

DESIGN ELEMENT	THEORETICAL GROUNDING	CoI PRESENCE
Korea Provocation	Garrison et al. (2000) — triggering event	Cognitive
Mirror Moment	Activates prior knowledge — Bloom (1956)	Cognitive + Learner
Context Capture	Hailey's suggestion — personalizes all responses	Teaching + Learner
Myth or Reality ×2	Hamilton (2021), Radianti (2020) — misconception check	Cognitive
Participation Metaphor	Sfard (1998) — reframes VR/AR as scaffold	Cognitive + Social
Equity Scenario	Lave & Wenger (1991), CAST (2018) — scenario-based	All four
Temperature Check	Hailey's suggestion — formative transition check	Social + Learner
Four Branching Paths	Problem-based learning — adaptive to named concern	Teaching + Learner
Postcard + Strategy	Metacognitive reflection + agency — Garrison (2009)	Learner
Curated Resources	Discipline-specific — contextual recommendation	Teaching



Learning Objectives — Bloom's Revised Taxonomy

1	DISTINGUISH	Between VR and AR and explain their distinct pedagogical affordances for higher education teaching
2	EVALUATE	The fit of VR/AR for a specific course or learning objective using a three-question evidence-informed feasibility framework
3	ANALYZE	How immersive technologies reshape participation in learning communities — including who benefits and who may be excluded
4	ASSESS	The equity and access implications of VR/AR adoption for their specific student population
5	REFLECT	On their own assumptions about immersive technology and articulate a personally grounded adoption decision



“The most important thing about immersive technology is not whether it is impressive. It is whether every student in your room gets to be genuinely immersed — not just the ones for whom the conditions already happen to be right.”

— Vera

THE CHATBOT

Seven-stage critical inquiry experience grounded in 9 research sources and 5 embedded assessments

THE THEORY

Sfard's participation metaphor — learning as joining a community, not just acquiring knowledge

THE QUESTION

VR/AR is hope or hype depending entirely on who gets to participate — and Vera helps educators find out

THANK YOU

GROUP 5 — LT 8440 | Spring 2026

Vimala

Hailey

A-Zahni

Chatbot: app.edcafe.ai/chatbots/69efdb894eb81875b9dac2e2

Instructor: Yin-Chan (Janet) Liao, Ph.D. | Georgia State University